

- 2 (a) A steel ball of diameter 5.0 mm is moving on a table with uniform acceleration a in a straight line with initial velocity u and final velocity v . The variation with time of the ball's velocity is shown in Fig. 2.1.

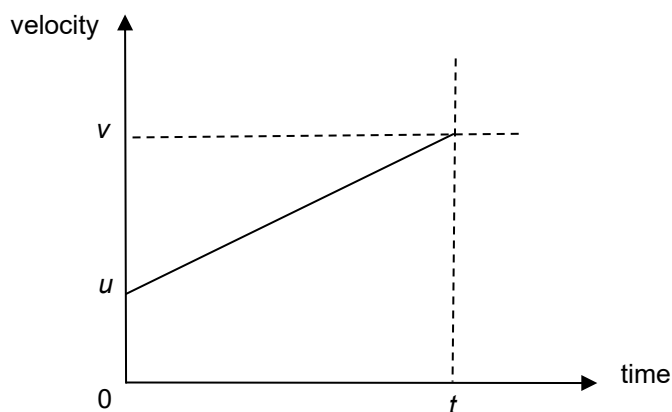


Fig. 2.1

Using the definition of acceleration and Fig. 2.1, show that $v^2 = u^2 + 2as$ where s is the displacement of the ball.

[2]

- (b) (i) The ball in (a) then falls off the table and lands on the floor. Draw on Fig. 2.2 the variation with time of the ball's vertical velocity (**V**) and horizontal velocity (**H**) from the instant the ball leaves the table to the instant before landing. Label the two lines as **V** and **H**.

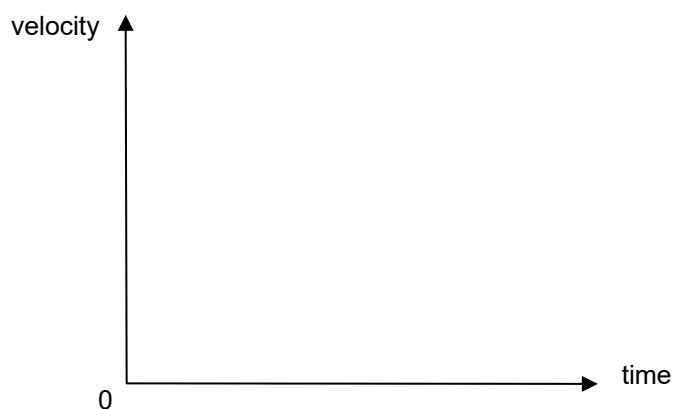


Fig. 2.2

[2]

- (ii) Draw on Fig. 2.3 the variation with horizontal distance of the object's vertical displacement. Consider downward displacement from the table top as positive.

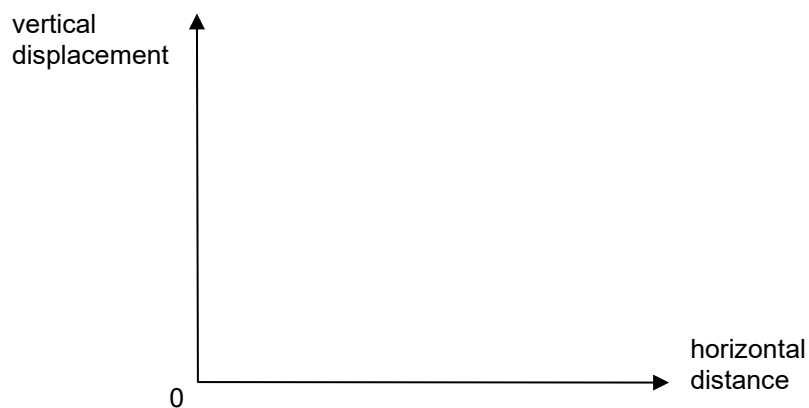


Fig. 2.3

[1]

- (iii) Suggest why air resistance may not be significant during the ball's motion.

.....

..... [1]

[Turn over